Note

- $A \cap \overline{A} = \phi$
- $A \cup \overline{A} = S$
- If A, B are mutually exclusive events, then $P(A \cup B) = P(A) + P(B)$
- P(Union of mutually exclusive events) $= \sum$(Probability of events)

Thorem 1

If A and B are two events associated with a random experiment, then prove that

(i) $P(A \cap \overline{B})$ = P(only A) $= P(A) - P(A \cap B)$

(ii) $P(\overline{A} \cap B)$ = P(only B) $= P(B) - P(A \cap B)$

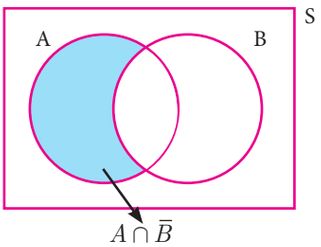


Fig. 8.7

Proof

(i) By Distributive property of sets,

1. $(A \cap B) \cup (A \cap \overline{B}) = A \cap (B \cup \overline{B}) = A \cap S = A$
2. $(A \cap B) \cap (A \cap \overline{B}) = A \cap (B \cap \overline{B}) = A \cap \phi = \phi$

Therefore, the events $A \cap B$ and $A \cap \overline{B}$ are mutually exclusive whose union is A.

Therefore, $P(A) = P\left[(A \cap B) \cup (A \cap \overline{B})\right]$

$P(A) = P(A \cap B) + P(A \cap \overline{B})$

Therefore, $P(A \cap \overline{B}) = P(A) - P(A \cap B)$

That is, $P(A \cap \overline{B})$ = P(only A) $= P(A) - P(A \cap B)$

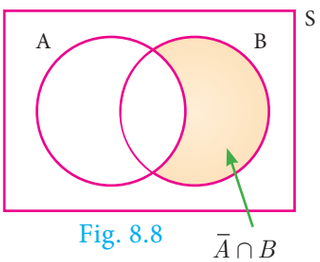


Fig. 8.8

(ii) By Distributive property of sets,

1. $(A \cap B) \cup (\overline{A} \cap B) = (A \cup \overline{A}) \cap B = S \cap B = B$
2. $(A \cap B) \cap (\overline{A} \cap B) = (A \cap \overline{A}) \cap B = \phi \cap B = \phi$

Therefore, the events $A \cap B$ and $\overline{A} \cap B$ are mutually exclusive whose union is B.

$P(B) = P\left[(A \cap B) \cup (\overline{A} \cap B)\right]$

$P(B) = P(A \cap B) + P(\overline{A} \cap B)$

Therefore, $P(\overline{A} \cap B) = P(B) - P(A \cap B)$

That is, $P(\overline{A} \cap B) = P$(only B) $= P(B) - P(A \cap B)$

Progress Check

1. P(only A)= _____.
2. $P(\overline{A} \cap B)$ = _____.
3. $A \cap B$ and $\overline{A} \cap B$ are _____ events.
4. $P(\overline{A} \cap \overline{B})$ = _____.
5. If A and B are mutually exclusive events then $P(A \cap B)$ = _____.
6. If $P(A \cap B) = 0.3$, $P(\overline{A} \cap B) = 0.45$ then $P(B)$ =_____.

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